



NITIN KUMAR

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LinkedIn — GitHub — GeeksForGeeks

Summary

Engineering student at NIT Calicut with strong foundations in software engineering, backend development and data structures. Experienced in building scalable applications and systems-oriented projects involving APIs, databases and performance optimization. Familiar with system design basics and exploring AI engineering and emerging technologies.

Technical Skills

Programming Languages: C++, JavaScript

Web Development: React.js, Node.js, Express.js, REST APIs

Databases: MongoDB, MySQL, Redis

AI & Generative AI (Basics): LLM Fundamentals, Prompt Engineering, LLM APIs, RAG Concepts, Embeddings Basics

System Design: API Design, Database Design, Caching, Scalability Basics

DevOps (Basics): Git, GitHub, Docker, Linux Basics

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems

Projects

Deep Packet Inspection

Network Security and Traffic Analysis Engine

Tech Stack: C++, TCP/IP, PCAP, Linux, Multithreading

- Developed a multi-threaded Deep Packet Inspection engine to analyse captured network traffic and extract protocol-level information.
- Implemented packet parsing for Ethernet, IPv4, TCP and UDP protocols with flow-based traffic analysis and packet classification.
- Designed efficient packet processing workflows using queues and worker threads for scalable network traffic handling.

CodeArena

Scalable Online Coding Platform for DSA Practice

Tech Stack: React.js, Node.js, Express.js, MongoDB, Redis, BullMQ, Kafka, SSE, Judge0 API, Gemini AI

- Built a scalable coding platform supporting DSA practice, automated code execution and submission evaluation.
- Designed asynchronous execution pipelines using BullMQ, Redis and background workers for efficient code processing.
- Integrated Kafka and Server-Sent Events (SSE) for real-time notifications and Gemini AI integration for coding guidance and problem explanations.

Experience

Web Development Intern – Unified Mentor

Live Demo

FerryFlow — Real-Time Ferry Operations & Passenger Management System

Tech Stack: React.js, Node.js, Express.js, MongoDB, Socket.IO, Redis

- Developed a full-stack ferry management platform with role-based workflows for passengers, staff and administrators.
- Designed REST APIs for scheduling, bookings, and passenger management, and implemented WebSocket communication for real-time operational updates.

Education

National Institute of Technology, Calicut

B.Tech. in Civil Engineering

Aug 2023 – Present

CGPA: 7.98

Inter College Pipra Nauranga, Khagaria

12th – BSEB

2021 – 2022

Percentage: 90%

Patliputra Central School, Parbatta, Khagaria, Bihar

10th – CBSE

2019 – 2020

Percentage: 87.2%

Achievements

- Solved **700+** Data Structures and Algorithms problems across GeeksForGeeks and LeetCode.
- Achieved **1700+ rating** on GeeksForGeeks competitive programming platform.
- Earned **SQL 50 Badge** on LeetCode.
- Selected as **Foundation For Excellence (FFE) Scholar and Mentee**.

Positions of Responsibility

- Volunteer, Student Guidance Cell (SGC), NIT Calicut
- Volunteer, National Service Scheme (NSS), NIT Calicut
- Senior Executive, Gaming Program Committee (GPC), Tathva'25
- Senior Executive, Workshop Committee, Ragam'26

Certifications

- Data Structures & Algorithms using C++ – Coders Army
- Backend Development – Udemy
- MERN Stack Development – Coursera

Interests

Exploring Emerging Technologies, Chess, Cricket, Badminton, Fitness Training